

Reflective Synthesis Paper

European AI Career Intelligence System (EACIS)

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1. Overview and Industry Context

The European AI job market is experiencing rapid structural change. The share of EU enterprises using AI rose from 8% in 2023 to 13.5% in 2024 (Eurostat, 2024), accelerating demand for qualified AI professionals across all sectors while the supply of talent remains constrained. Candidates entering or transitioning into the AI industry face three major problems: they do not know which roles their current skills qualify them for, what roles are available on the European job market or how to bridge the gap between their existing profile and their target role. Employers face other aspects of these problems: job titles are inconsistently defined regarding actual skill demand and the absence of a standardised European AI role taxonomy makes matching candidates to roles inefficient. Platforms such as Jobright.ai have addressed this problem in the United States through AI-assisted matching, automated application tools and a 24/7 career copilot - demonstrating clear product-market fit at scale. However, no equivalent platform exists for the European market, which operates under distinct regulatory constraints (the EU AI Act, GDPR), has a multilingual job landscape and requires salary benchmarking that US-derived data cannot provide. Meanwhile, platforms with richer data - LinkedIn, Indeed, Glassdoor - prohibit automated access to their job listings in their Terms of Service, making Adzuna, a British job aggregator operating in 19 countries, the most viable open data source for European labour market research. This project addresses this gap by designing and implementing the European AI Career Intelligence System (EACIS) - a multi-layer AI platform that combines labour market analytics, semantic role understanding and agentic career guidance into a single integrated system. EACIS is candidate-facing, transparent by design and built with EU AI Act compliance as an architectural requirement.

2. Integration Rationale

EACIS integrates all six prior capstone projects, each contributing a distinct and necessary layer. The integration is not additive but compositional - removing any single layer would break the system's ability to produce its end-to-end output.

Project 1 provides the data infrastructure: the Adzuna API pipeline collects and enriches job postings across six European countries and the 500-character API truncation problem - discovered and solved in Project 1 - shaped every downstream design decision, from the description enrichment step to the choice of DistilBERT over bag-of-words methods that would have been severely limited by short descriptions. Project 2 contributes the salary intelligence layer: the ANOVA finding that remote work mode significantly affects salary ($F=4.56$, $p=0.0107$) (Lusa et al., 2024) and the EUR normalisation via ECB exchange rates provide the compensation benchmarks embedded in every career report. The statistical rigour of Project 2's initial data analysis framework - systematic data screening, outlier documentation and transparent preprocessing - also established the methodological standards carried forward into all subsequent work.

Project 3 supplies the skill taxonomy and role archetypes that form the semantic core of the matching engine. The discovery that AI job titles are unreliable proxies for skill profiles - AI Consulting and AI Product or Project Management roles failed to form distinct skill-based clusters, distributing across engineering archetypes - directly motivated the geographic expansion of Project 4 and validated the fundamental premise of EACIS: that skill-based matching over label-based categorisation is necessary for accurate career guidance.

Project 4 deepened this representation by replacing keyword vectors with DistilBERT semantic embeddings, and the empirical comparison of TF-IDF versus DistilBERT clustering provided the evidence base for the Transformer-based approach used in production (Sanh et al., 2019). The FAISS index of 1,088 encoded job descriptions from Project 4 became the retrieval backbone of the RAG component (Lewis et al., 2020), grounding every recommendation in real market evidence.

Project 5's contribution is more nuanced: GPT-2 fine-tuning was tested and rejected after hallucinations were observed, but the experiment produced the three-tier fit routing logic, the cosine similarity thresholds and the structured template format that replaced it. The failure of GPT-2 was therefore a productive engineering outcome - it established an evidence-based upper bound on what small language models can reliably do in this domain and provided the design rationale for the template-based generation approach used in the production system.

Project 6 synthesises all components into the ReAct agent (Yao et al., 2022), providing the reasoning loop, state management and transparent decision trace that make the system interpretable and auditable.

3. System Design and Technical Decisions

EACIS is structured as a six-layer pipeline with a strict dependency chain. The offline pipeline - data collection, skill extraction, archetype clustering, DistilBERT encoding, FAISS index construction and salary pre-computation - runs once to build the system artefacts. The online pipeline runs at query time, executing six sequential tool calls within the ReAct agent: skill extraction, archetype matching, RAG retrieval, recommendation generation, live job fetching and salary lookup.

Three technical decisions require explicit justification. First, the choice of DistilBERT over multilingual models such as mBERT or XLM-RoBERTa. This was deliberate: over 85% of collected job postings are in English even for DACH markets, as AI roles in Germany and Austria are frequently advertised in English to attract international candidates. The computational cost of multilingual Transformer models - significantly larger and slower on CPU - would have made the FAISS index construction impractical within academic hardware constraints. Multilingual encoding is explicitly identified as the appropriate production upgrade, not an oversight. Second, the replacement of GPT-2 generation with deterministic templates. As Bender et al.(2021) warn, language models trained on web-scraped text can produce plausible-sounding but factually incorrect outputs, and this risk is particularly acute in career guidance where candidates may act on the system's recommendations. The template approach sacrifices output diversity for reliability - a tradeoff that is explicitly appropriate for a system where accuracy matters more than generative creativity. Third, the single-agent sequential architecture over multi-agent parallelism. Sequential execution simplifies state management, makes the reasoning trace fully legible and reduces the risk of inter-agent coordination failures. The latency tradeoff is acceptable for a candidate-facing orientation tool where session time is not a primary constraint.

4. Ethical Considerations and Responsible AI

The EU AI Act classifies recruitment and HR systems as high-risk AI applications under Annex III, with core compliance requirements - including documentation, human oversight and bias auditing - becoming enforceable in August 2026. EACIS is designed to anticipate these requirements rather than retrofit them.

The primary ethical risk is misuse: the candidate-facing guidance system could be repurposed as a recruiter-facing screening tool. If archetype similarity scores were used to filter or rank applicants, the system would encode and amplify the biases present in its training data - job postings predominantly from Germany and the UK that reflect those markets' hiring norms, language and role structures. Candidates from non-English-speaking backgrounds, non-traditional career paths or markets not well-represented in the Adzuna index could be systematically disadvantaged even when

they possess equivalent competencies expressed in different terminology. Jobin et al. (2019) identify non-maleficence, transparency and human oversight as the most consistently cited principles in global AI ethics frameworks - all three are directly implicated here.

A second concern is the system's production of confident-sounding recommendations without uncertainty quantification. A candidate receiving a no-fit response based on a narrow 50-skill keyword match may be discouraged from pursuing AI roles despite having relevant experience outside the taxonomy. The current implementation addresses transparency through the reasoning trace - every step of the agent's decision process is visible to the user - but does not yet present confidence intervals or multiple archetype scores to the candidate. Wachter et al. (2017) argue that automated decision systems must provide meaningful explanations to affected individuals; the reasoning trace is a first step toward this standard, but fuller uncertainty communication remains a required improvement.

The system's geographic data bias - Germany and the UK account for over 80% of collected postings - is documented explicitly in every output and in the system's responsible use documentation. The salary ranges are labelled as DACH-specific rather than pan-European and the archetype definitions are presented as market-derived rather than universal. These design choices do not eliminate the bias but make it legible, which is the minimum standard for responsible deployment in the current regulatory environment.

5. Evaluation and Reflection

The four-profile EACIS demonstration produced results that both validated the system's core hypothesis and revealed its key limitations. The automotive engineer result - a strong archetype match to Deep Learning & AI Research from two skills alone - confirmed that skill-based matching surfaces relevant career pathways regardless of a candidate's industry of origin, which is the system's fundamental value proposition over conventional keyword job search. All seven integration layers were exercised across all four profiles, validating the pipeline's end-to-end functionality as an integrated system. The most significant limitation is the 50-skill taxonomy. The automotive engineer profile detected only 2 of the candidate's many relevant skills - terms like 'embedded systems', 'sensor data', 'image processing pipelines' and 'C++' are absent from the taxonomy, causing the system to report 25% archetype coverage when the true competency alignment is likely much higher. This limitation propagates through every downstream component and is the single highest-priority improvement for any production deployment. A named entity recognition model fine-tuned on resume text would address this without sacrificing the interpretability of the keyword approach.

Integrating across six projects produced insights that no single project could have revealed.

The progression from keyword clustering (Project 3) to semantic embeddings (Project 4) demonstrated empirically that Transformer representations produce more coherent role archetypes - a finding that directly justified the DistilBERT choice in the production system.

The GPT-2 failure in Project 5 was frustrating in isolation but informative in context: it established that reliable domain-specific text generation requires either larger models or stronger constraints and the template solution that replaced it is actually more appropriate for a high-stakes guidance application. These are the kinds of insights that only emerge from end-to-end system integration.

6. Professional Relevance

EACIS demonstrates several competencies directly relevant to industry AI roles. First, end-to-end pipeline ownership: the system spans data collection, statistical analysis, machine learning, deep learning, generative AI and agentic orchestration - a breadth of technical integration that mirrors the requirements of senior ML engineering and AI product roles.

Second, evidence-based design decision making: every architectural choice - template generation over GPT-2, DistilBERT over multilingual models, single-agent over multi-agent - is justified by empirical evidence from prior experiments rather than preference or convention.

Third, regulatory awareness: designing for EU AI Act compliance from the outset, rather than treating ethics as a documentation exercise, reflects the standard expected of AI practitioners working in the European market from August 2026 onward.

Fourth, honest failure analysis: the documented GPT-2 failure, the taxonomy coverage limitations and the geographic data bias are presented as engineering findings rather than concealed - a professional practice that distinguishes reliable practitioners from those who overstate system capabilities.

The full project series - from Adzuna data collection to an agentic career guidance system - constitutes a coherent, end-to-end AI product prototype. Positioned against Jobright.ai as a European-focused, transparency-first alternative, it demonstrates not only technical capability but the ability to frame technical work within a realistic industry context, identify a genuine market gap and make principled architectural decisions under real constraints. These are the capabilities that advanced industry and research roles require.

7. References

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